

National University of Computer and Emerging Sciences

Feedback Control Systems (EE3004)

Date: June 6th 2024

Course Instructor

Dr. Omer Saleem (BEE-6A)

Final Exam

Total Time: 3 Hours

Total Marks: 100

Total Questions: 05

Semester: SP-2024

Campus: Lahore

Dept: Electrical Engg.

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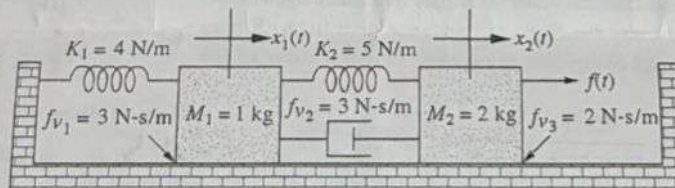
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CLO # 1: Develop the mathematical model of a physical system.

Q1:

[10 + 10 marks]

- a. Find the state space representation of the following mechanical system.



$$X(s)(Ms^2 + FVs + K) = F(s)$$

$$M\ddot{x}(t) + F_v\dot{x}(t) + Kx(t) = F(t)$$

$$z_1 = \dot{x}(t)$$

$$z_2 = \ddot{x}(t) = \dot{z}_1$$

$$\dot{z}_2 = \ddot{x}(t) = \ddot{z}_1$$

- b. Convert the following state space representation into a transfer function of the form, $Y(s)/U(s)$. Hence, find the expression of output, $y(t)$, in time-domain if the input signal, $x(t)$, is a unit-step signal.

$$\frac{Y(s)}{U(s)} = \frac{4 + s}{s^2 + 5s + 4}$$

$$\dot{x}(t) = \begin{bmatrix} 0 & 2 \\ -2 & -5 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 2 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \end{bmatrix} u(t)$$

$$C(sI - A)^{-1}B + D$$

$$\frac{y(t)}{x(t)}$$

CLO # 2: Analyze the transient and steady state response of a feedback system.

Q2: The system shown below has its transient response altered by adding a tachometer.

[20 marks]

- Reduce the block diagram to find the system's open-loop transfer function $G(s)$ as well as overall transfer function $T(s)$.
- Design K and K_2 in the system to yield a damping ratio of 0.69 and a natural frequency of 10.0 rad/s.
- What is the steady-state error of this closed-loop system?
- What modification is needed in the pre-amplifier K to ensure the steady-state error becomes zero?

$$T(s) = \frac{s}{s+1}$$

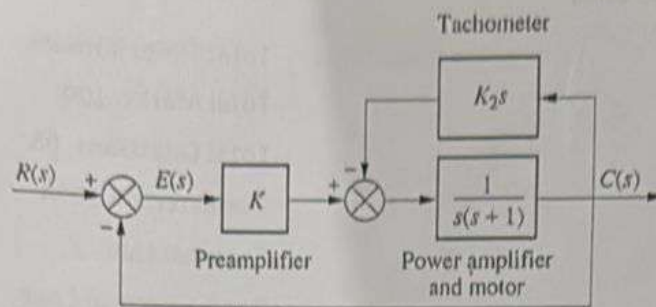
$$s^0 (s^2 + 13.8s + 100)$$

Page 1 of 3

$$\text{Type 0} \rightarrow \frac{1}{1-10}$$

$$s^2 + 13.8s$$

$$s(s + 13)$$

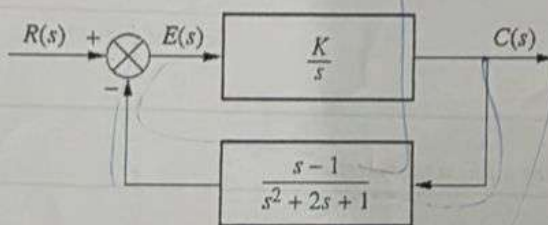


CLO # 3: Analyze the stability of a system.

Q3:

[20 marks]

- For the unity feedback system shown below:
 - Find the closed-loop transfer function, $T(s)$, of the system.
 - Hence, use $T(s)$ to find the range of K for stability.



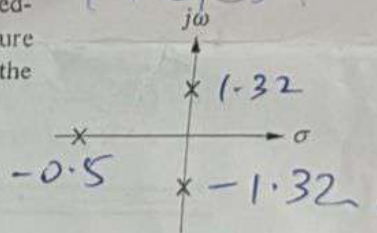
$$T(s) = \frac{G(s)}{1 + G(s)H(s)}$$

$$G(s) = \frac{K}{s}$$

$$H(s) = \frac{s-1}{s^2+2s+1}$$

- Use Routh Hurwitz criteria to find the value of K that will place the closed-loop poles of the following unity feedback system as shown in the Figure here; one pole on real-axis and two poles on $j\omega$ -axis. Find the value of the poles as well.

$$G(s) = \frac{Ks^2(s+1)}{s^3 + s^2 + 2s + K}$$



$$-\frac{2}{3} < K < 0$$

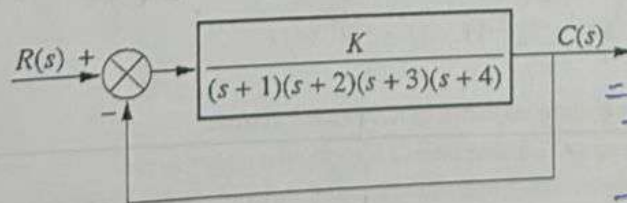
$$\pm 1.4j$$

CLO # 4: Analyze a given system in frequency domain.

Q4:

[20 marks]

- For the system shown in Figure below,



$$\sigma = \frac{\sum p - \sum z}{\#p - \#z}$$

$$= \frac{(-1-2-3-4) - 0}{4 - 0}$$

$$= -5/2 = -2.5$$

$$\theta_\sigma = \frac{(2n+1)180}{\#p - \#z}$$

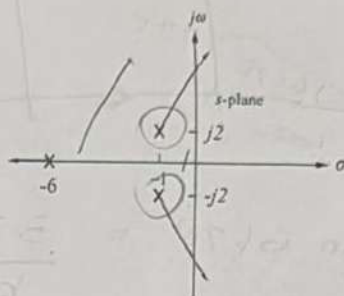
$$n=1, \frac{(2 \times 1 + 1)180}{4 - 0}$$

$$n=2, \frac{(2 \times 2 + 1)180}{4 - 0}$$

- Find the asymptotes of the root-locus.
- Find the break-away and break-in points (if any) of the root-locus.
- The crossings on the $j\omega$ -axis of the root-locus.
- Hence, sketch the root-locus.

2, 4

- b) The root locus of a system, $G(s)$, is shown here:
- Find the open-loop transfer function, $G(s)$, of the system.
 - Hence, find the value of associated gain K for which the closed-loop function will yield a closed-loop pole on the real-axis at $s = -8$.



CLO # 5: Design a closed-loop feedback controller

Q5: The open loop transfer function of an un-compensated unity feedback system is given below. [20 marks]

$$KG(s)H(s) = \frac{K}{(s+1)(s+4)}$$

The un-compensated system has a dominant pole at $-2.5 + j1.875$. The damping ratio of the un-compensated system is 0.8 .

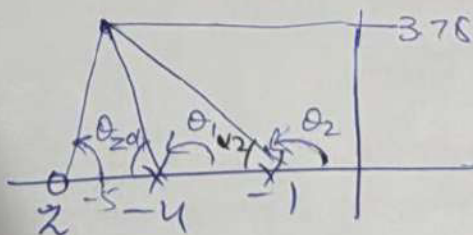
- Find the un-compensated system's settling-time and peak-time.
- It is desired to compensate the system to decrease its settling-time by a factor of 2.0 and steady-state error to zero without affecting the damping ratio. Find the new settling-time, peak-time, and the dominant pole location to satisfy the required performance metrics.
- Hence, design a PID controller to compensate the system to achieve the performance in part (ii). Write the overall transfer function of the system including plant and controller pole/zeros and gain.

$$(s+7) \left(\frac{s+0.01}{s} \right)$$

$$\frac{(s+7)(s+0.01)}{s} = \frac{s^2 + 0.01s + 7s + 0.07}{s} = \frac{s^2 + 7.01s + 0.07}{s}$$

$$b = 5.778$$

$$= 2.352$$



$$\theta_2 \rightarrow \tan \alpha_2 = \frac{3.75}{4} = 43.15^\circ \Rightarrow 180 - 43.15^\circ$$

$$\theta_2 = 136.85^\circ$$

$$\theta_1 = \tan \alpha_1 = \frac{3.75}{1} = 75.07^\circ \Rightarrow 180 - 75.07^\circ$$

$$\theta_1 = 104.93^\circ$$

$$+\theta_z - \theta_1 - \theta_2 = 180$$

$$\theta_z = 421.78 - 360$$

$$\theta_z = 61.78^\circ$$